Notes: Kowalski (RES, 2023)

Behaviour within a Clinical Trial and Implications for Mammography Guidelines

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1 Big picture

- **Question.** Do current U.S. mammography guidelines for women in their 40s induce mammograms among women who are most likely to benefit, or among women who are more likely to be overdiagnosed?
- **Policy setting.** The USPSTF weakened guidelines in 2009 and left screening for women aged 40–49 to individual choice with physicians.
- **Clinical trial.** The paper studies the Canadian National Breast Screening Study (CNBSS), which combines random assignment, imperfect mammography take-up, and long-run follow-up.
- **Core empirical move.** Random assignment instruments for actual mammography, and imperfect compliance separates always takers, compliers, and never takers.
- **Main finding.** Women most likely to receive mammograms are healthier and higher socioeconomic status, but have much higher overdiagnosis: at least 36% of always-taker cancers are overdiagnosed.
- **Policy implication.** Current U.S. guidelines may encourage mammograms among healthier women who are more likely to be overdiagnosed by them.

2 Incremental contribution of the paper

- **Beyond average trial effects.** Existing mammography evidence focuses on average effects; this paper asks how effects vary with mammography behaviour.
- **Behaviour-within-trial framework.** Noncompliance is used constructively to identify latent groups that map to guideline regimes.
- **Selection heterogeneity test.** Under LATE assumptions alone, the paper tests whether screening-prone women differ in untreated outcomes and covariates.
- **Treatment-effect heterogeneity.** With a weak monotonicity assumption on untreated outcomes, the paper bounds always-taker treatment effects and compares them to complier LATE.
- **New substantive result.** Prior CNBSS evidence supports overdiagnosis on average; this paper shows overdiagnosis is especially high among women most likely to screen under weak guidelines.

Interpretation. The paper contributes both a general method for studying policy targeting inside clinical trials and a specific mammography-guideline result.

3 Data and measurement

3.1 CNBSS trial and randomization

The CNBSS enrolled almost 90,000 Canadian women aged 40–59 between 1980 and 1985. All women received a clinical breast examination at enrolment before individual-level random assignment to intervention or control, stratified by age and study centre. Intervention-arm women received annual mammograms and clinical breast examinations during the active study period; control-arm women aged 40–49 received usual care.

Interpretation. The trial is not a simple full-compliance experiment. Noncompliance creates the behavioural variation used in the paper.

3.2 Main sample

The main sample is women aged 40–49 at enrolment, excluding women with family history of breast cancer, previous breast cancer, other breast disease, symptoms, nurse abnormalities, or referral for review. The main sample has 19,505 women.

Interpretation. These exclusions make the sample closer to asymptomatic women without known elevated risk, the population targeted by screening guidelines.

3.3 Treatment, instrument, and outcomes

- Treatment $D = 1$ if the participant receives a mammogram in at least one active-study year after the enrolment year.
- Instrument $Z = 1$ if assigned to intervention.
- First-stage probabilities are $p_C = P(D = 1|Z = 0) = 0.19$ and $p_I = P(D = 1|Z = 1) = 0.95$.
- Main outcomes are 20-year breast cancer incidence and all-cause mortality.
- Additional cancer characteristics include tumour size, invasive status, and mastectomy among cancers detected during the active period.

Interpretation. Long follow-up is essential because overdiagnosis is a persistent excess of diagnosed cancers, not merely short-run lead time.

4 Models and empirical specifications

4.1 Potential outcomes and LATE setup

$$Y = Y_U + (Y_T - Y_U)D.$$

Y_T is the outcome under mammography and Y_U is the untreated outcome.

Interpretation. For breast cancer incidence, a positive treatment effect over long follow-up can indicate overdiagnosis.

4.2 First stage and latent groups

$$\text{Always takers} : p \in [0, p_C], \qquad \text{Compliers} : p \in (p_C, p_I], \qquad \text{Never takers} : p \in (p_I, 1].$$

Interpretation. Always takers are most analogous to women who screen under weak voluntary guidelines; compliers are induced by stronger screening encouragement.

4.3 Latent group outcomes

$$E[Y_T|AT] = E[Y|D = 1, Z = 0], \qquad E[Y_U|NT] = E[Y|D = 0, Z = 1],$$
$$E[Y_T|C] = \frac{p_I E[Y|D = 1, Z = 1] - p_C E[Y|D = 1, Z = 0]}{p_I - p_C},$$
$$E[Y_U|C] = \frac{(1 - p_C)E[Y|D = 0, Z = 0] - (1 - p_I)E[Y|D = 0, Z = 1]}{p_I - p_C}.$$

Interpretation. These formulas decompose observed treatment-assignment cells into always-taker, complier, and never-taker outcomes.

4.4 Untreated outcome test

$$H_0 : E[Y_U|C] = E[Y_U|NT].$$

Interpretation. Rejection means women with different screening propensities have different untreated health outcomes.

4.5 M.1 weak monotonicity and always-taker lower bound

For $p_1 < p_2$, M.1 assumes $E[Y_U|p_1]$ is weakly monotone in p. Given the observed direction of selection, it yields an upper bound on always takers' untreated cancer incidence and thus a lower bound on their treatment effect.

Interpretation. This is the extra assumption needed to move from selection heterogeneity to treatment-effect heterogeneity for always takers.

5 Identification strategy

- **Random assignment.** Z is randomly assigned, so it can instrument actual mammography.
- **Monotonicity.** Intervention assignment weakly increases mammography; no defiers.
- **Noncompliance as information.** Take-up and assignment partition women into always takers, compliers, and never takers.
- **Selection heterogeneity.** The untreated outcome test compares compliers and never takers using outcomes observed without mammography and requires only LATE assumptions.
- **Treatment heterogeneity.** M.1 bounds always-taker treatment effects and allows comparison with complier LATE.
- **Validation.** Baseline covariates and cancer characteristics check whether screening-prone women appear healthier and whether detected cancers look less severe.
- **Robustness.** Results are examined across outcomes, samples, mammography definitions, and follow-up lengths.

Interpretation. The strongest identification result is selection heterogeneity. Treatment-effect heterogeneity requires M.1, but the assumption is intentionally weak and aligned with the observed direction of selection.

6 Tables and figures

6.1 Figure 1: Ranges of the fraction treated p for always takers, compliers, and never takers

Read: Figure 1 maps $p_C = 0.19$ and $p_I = 0.95$ into always takers, compliers, and never takers.

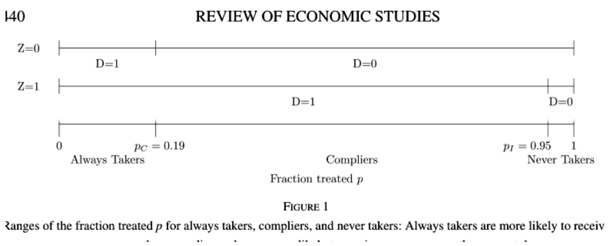


Figure 1: Ranges of the fraction treated p.

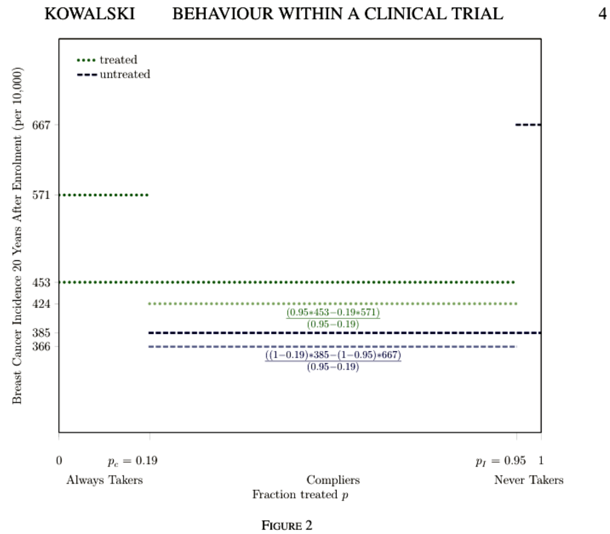


Figure 2: Derivation of latent-group average outcomes.

Interpretation. The first stage converts noncompliance into policy-relevant groups.

6.2 Figure 2: Derivation of average breast cancer incidence for always takers, compliers, and never takers

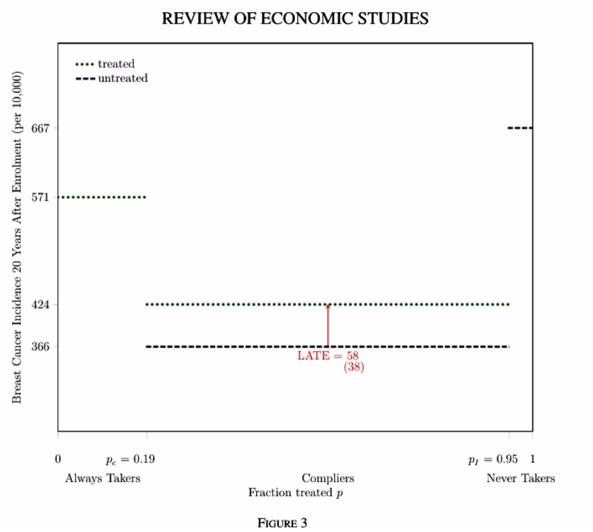
Read: Figure 2 identifies treated and untreated outcomes for latent groups; the complier LATE is $424 - 366 = 58$ breast cancers per 10,000.

Interpretation. This figure is the core decomposition linking observed cells to latent potential outcomes.

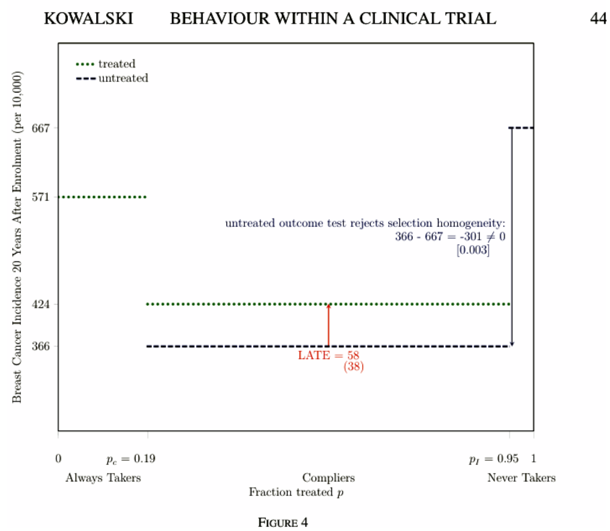
6.3 Figure 3 and Figure 4: Average breast cancer incidence and untreated outcome test

Read: Figure 4 shows untreated compliers at 366 versus never takers at 667 breast cancers per 10,000, rejecting selection homogeneity.

Interpretation. Women more likely to receive mammograms are healthier in terms of untreated breast cancer incidence.



(a) Figure 3



(b) Figure 4

Breast cancer incidence and untreated outcome test.

6.4 Table 1: Baseline covariates corroborate selection heterogeneity

Read: Table 1 shows that always takers and compliers tend to have higher socioeconomic status and healthier baseline behaviours than never takers.

Interpretation. Voluntary screening is positively selected even before treatment begins.

TABLE 1

Baseline covariates corroborate selection heterogeneity: Women more likely to receive mammograms have highe socioeconomic status and are more likely to practice other health behaviours seen as beneficial

	Means			Difference in Mean:	
	(1) Always Takers	(2) Compliers	(3) Never Takers	(1)–(2)	(2)–(:
Baseline Socioeconomic Status					
In work force	0.65	0.64	0.65	0.02	-0.02
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Age at first birth	24.28	23.98	23.57	0.30	0.41
	(0.11)	(0.04)	(0.21)	(0.13)	(0.22
No live birth	0.16	0.15	0.13	0.01	0.01
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Married	0.80	0.81	0.75	-0.01	0.06
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Husband in work force if alive	0.81	0.81	0.76	-0.00	0.05
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Baseline Health Behaviour					
Non-Smoker	0.78	0.75	0.63	0.03	0.12
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Body Mass Index	23.87	24.42	24.48	-0.56	-0.06
	(0.10)	(0.04)	(0.22)	(0.11)	(0.23
Used oral contraception	0.74	0.71	0.67	0.03	0.04
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Used estrogen	0.13	0.13	0.15	-0.00	-0.02
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Any mammograms prior to enrolment	0.23	0.13	0.13	0.10	-0.00
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02
Practiced breast self-examination	0.47	0.44	0.38	0.03	0.06
	(0.01)	(0.00)	(0.02)	(0.01)	(0.02

Notes: Bootstrapped standard errors are under point estimates in parentheses. Each line of the table reports stati on a different baseline covariate X. The treatment D is mammography, which is equal to one if a participant rece a mammogram in at least one year during the active study period after the enrolment year. The instrument Z is e

Table 1: Baseline covariates and selection heterogeneity.

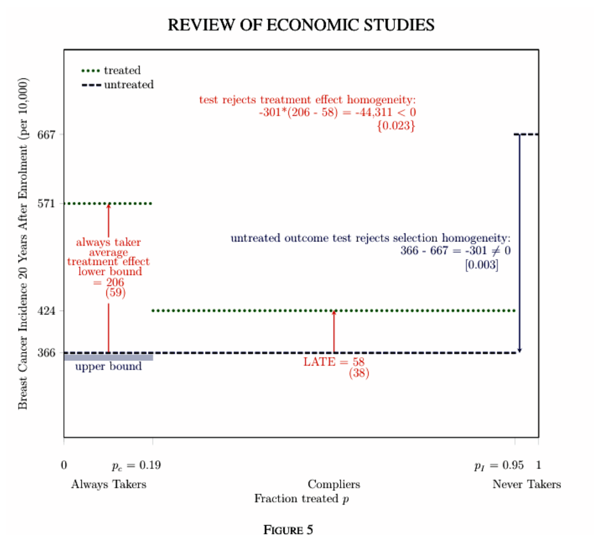


Figure 5: Treatment-effect heterogeneity and overdiagnosis.

6.5 Figure 5: Treatment effect homogeneity test and always-taker overdiagnosis

Read: Figure 5 gives an always-taker treatment-effect lower bound of 206 versus complier LATE 58; at least 36% of always-taker cancers are overdiagnosed.

Interpretation. The women most likely to screen under weak guidelines have the largest overdiagnosis lower bound.

6.6 Table 2: Breast cancer characteristics

Read: Always-taker cancers are smaller on average and less invasive; mastectomy is more common among always takers conditional on surgery.

Interpretation. The cancer characteristics corroborate selection and treatment-effect heterogeneity and suggest possible collateral harms.

TABLE 2

Suggestive evidence that women more likely to receive mammograms have breast cancers that are smaller and le invasive and undergo more aggressive procedures

	Means		Difference in M
	(1) Always Takers	(2) Treated Compliers	(1)–(2)
Tumour Size Among Breast Cancers (in mm)	13	18	-5
	(2)	(3)	(4)
Share of Invasive Breast Cancer Among	73	75	-2
Breast Cancers (%)	(9)	(7)	(13)
Share of Mastectomy Among Breast Cancers	45	23	22
with Mastectomy or Lumpectomy (%)	(9)	(7)	(14)

Notes: Bootstrapped standard errors are under point estimates in parentheses. The treatment D is mammography, wl is equal to one if a participant receives a mammogram in at least one year during the active study period after enrolment year. Each outcome Y is restricted to the years for which treatment is defined during the active study per Lumpectomy is a procedure that involves partial removal of a breast, and mastectomy is a more aggressive proce that involves complete removal of a breast. The instrument Z is equal to one if a participant is assigned to intervent The main analysis sample includes women aged 40–49 at enrolment and excludes those who report any breast canc their family, any previous breast cancer diagnosis, any other breast disease, or any symptoms, as well as those for wl a nurse found abnormalities or referred them for review.

Table 2: Breast cancer characteristics and procedures.

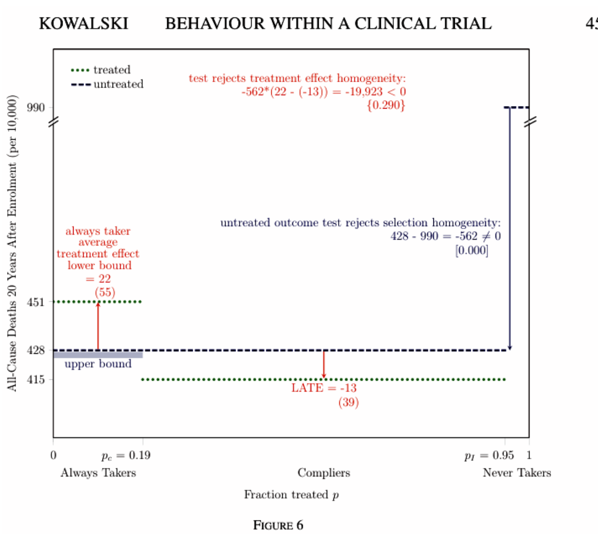


Figure 6: All-cause mortality.

6.7 Figure 6: All-cause mortality

Read: Untreated compliers have 428 deaths per 10,000 versus 990 for never takers; treatment-effect heterogeneity is directionally consistent but not statistically significant.

Interpretation. Selection heterogeneity is strong even for mortality; treatment-effect evidence is weaker.

6.8 Table 3: Robustness to alternative outcomes, samples, and mammography definitions

Read: The selection result is robust across outcomes, samples, and treatment definitions; treatment-effect heterogeneity is strongest for the main breast cancer specification.

Interpretation. The main finding is not a narrow artifact of sample selection or treatment definition.

TABLE 3

Summary of findings depicted in Figure 5 and robustness to alternative outcomes, sample restrictions, and definitions of mammography

	N	(1) Untreated Outcome Test Rejects Selection Homogeneity	(2) Always Taker Average Treatment Effect Lower Bound	(3) Local Average Treatment Effect LATE	(4) Test Rejects Treatment Effect Homogeneity $(1)^*((2)-(3))<0$
Main Specification					
Outcome is breast cancer incidence, sample is main analysis sample, treatment is defined as mammogram in at least one active study period after enrolment					
Breast cancer incidence	19,505	-301	206	58	-44,311
		[0.003]	(59)	(38)	{0.023}
Alternative Outcomes					
All-cause mortality	19,505	-562	22	-13	-19,923
		[0.000]	(55)	(39)	{0.290}
Alternative Sample Restrictions					
All excluded participants aged 40–49 at enrolment	30,925	-1,237	309	79	-284,634
		[0.000]	(45)	(44)	{0.000}
All participants aged 40-49 at enrolment	50,430	-826	298	69	-189,397
		[0.000]	(36)	(30)	{0.000}
All participants aged 50-59 at enrolment	39,405	-1,555	419	39	-591,037
		[0.000]	(53)	(34)	{0.000}
All participants	89,835	-1,156	332	55	-319,660
		[0.000]	(30)	(22)	{0.000}
Alternative Definitions of Mammography					
At least two active study period years after enrolment	19,505	-341	239	54	-63,347
		[0.000]	(90)	(35)	{0.019}
At least three active study period years after enrolment	19,505	-330	167	55	-36,927
		[0.000]	(142)	(36)	{0.206}
All active study period years after enrolment	19,505	-178	158	64	-16,656
		[0.005]	(181)	(42)	{0.312}

Notes: Bootstrapped standard errors are under point estimates in parentheses, two-tailed bootstrapped p-values are under test statistics in brackets, and one-tailed bootstrapped p-values are under test statistics in curly braces. Some p-values are zero if the test rejects the null hypothesis in all 1,000 bootstrap replications. Each outcome Y is measured 20 years after enrolment per 10,000 participants for all participants, based on initial occurrence and the exact calendar date of enrolment. In the main specification, the treatment D is mammography, which is equal to one if a participant receives a mammogram in at least 1 year during the active study period after the enrolment year. The instrument Z is equal to one if a participant is assigned to intervention. The main analysis sample includes women aged 40–49 at enrolment and excludes those who report any breast cancer in their family, any previous breast cancer diagnosis, any other breast disease, or any symptoms, as well as those for whom a nurse found abnormalities or referred them for review. Some differences between statistics might not appear internally consistent because of rounding.

Table 3: Robustness summary.

TABLE 4

Summary of findings depicted in Figure 5 and robustness to alternative follow-up lengths

Years Since Enrolment	N	(1) Untreated Outcome Test Rejects Selection Homogeneity	(2) Always Taker Average Treatment Effect Lower Bound	(3) Local Average Treatment Effect LATE	(4) Test Reject Treatment Ef Homogenei $(1)^*((2)-(3)$
Main specification: 20	19,505	-301	206	58	-44,311
		[0.003]	(59)	(38)	{0.023}
19	19,505	-269	196	52	-38,565
		[0.013]	(58)	(37)	{0.023}
18	19,505	-311	210	54	-48,503
		[0.000]	(56)	(35)	{0.010}
17	19,505	-322	214	49	-52,975
		[0.000]	(55)	(34)	{0.005}
16	19,505	-342	232	56	-60,245
		[0.000]	(54)	(32)	{0.003}
15	19,505	-381	211	84	-48,650
		[0.000]	(50)	(31)	{0.015}
14	19,505	-404	201	80	-49,046
		[0.000]	(49)	(29)	{0.020}
13	19,505	-431	223	75	-63,808
		[0.000]	(48)	(28)	{0.007}
12	19,505	-443	191	64	-56,156
		[0.000]	(44)	(27)	{0.010}
11	19,505	-423	195	55	-59,084
		[0.000]	(43)	(25)	{0.004}
10	19,505	-419	200	47	-64,017
		[0.000]	(42)	(23)	{0.000}
9	19,505	-413	192	34	-64,955
		[0.000]	(40)	(22)	{0.000}
8	19,505	-409	175	35	-57,386
		[0.000]	(37)	(21)	{0.000}
7	19,505	-393	177	46	-51,740
		[0.000]	(35)	(18)	{0.000}
6	19,505	-412	185	50	-55,761
		[0.000]	(33)	(17)	{0.000}
5	19,505	-382	180	45	-51,581
		[0.000]	(32)	(15)	{0.000}
4	19,505	-393	152	46	-41,568
		[0.000]	(29)	(13)	{0.003}
3	19,505	-354	104	37	-23,679
		[0.000]	(23)	(11)	{0.012}
2	19,505	-337	63	25	-12,632
		[0.000]	(18)	(9)	{0.030}
1	19,505	-342	35	20	-5,194
		[0.000]	(11)	(6)	{0.097}

Notes: Bootstrapped standard errors are under point estimates in parentheses, two-tailed bootstrapped p-values are u test statistics in brackets, and one-tailed bootstrapped p-values are under test statistics in curly braces. Some p-values zero if the test rejects the null hypothesis in all 1,000 bootstrap replications. The outcome Y is breast cancer incide measured at various years since enrolment for all participants, based on initial diagnosis and the exact calendar dat enrolment. The treatment D is mammography, which is equal to one if a participant receives a mammogram in at lea year during the active study period after the enrolment year. The instrument Z is equal to one if a participant is assig to intervention. The main analysis sample includes women aged 40–49 at enrolment and excludes those who report breast cancer in their family, any previous breast cancer diagnosis, any other breast disease, or any symptoms, as we

Table 4: Alternative follow-up lengths.

6.9 Table 4: Robustness to alternative follow-up lengths

Read: The untreated outcome test is negative and significant at each follow-up length; LATE remains positive across horizons.

Interpretation. The long follow-up supports an overdiagnosis interpretation rather than merely temporary lead time.

7 Interpretive synthesis

- Screening-prone women are healthier but are more likely to be overdiagnosed.
- The paper shows why voluntary guideline regimes may not target the women with the largest net benefit.
- Noncompliance in a randomized clinical trial becomes a source of policy-relevant heterogeneity.
- Modern technology may increase overdiagnosis by detecting smaller abnormalities, while modern treatment could reduce some collateral harms.

8 Short summary

- The paper studies mammography behaviour in the CNBSS.
- Always takers, compliers, and never takers are defined using random assignment and take-up.
- Women more likely to screen are healthier and higher socioeconomic status.
- Always takers have at least 3.5 times higher 20-year overdiagnosis than compliers.
- At least 36% of always-taker cancers are overdiagnosed.
- Current U.S. guidelines may encourage screening among women more likely to be overdiagnosed.

9 Conclusion

9.1 Core conclusion

Behaviour within a clinical trial reveals that voluntary mammography is not targeted to the women most likely to benefit. Women most likely to receive mammograms are healthier and more likely to be overdiagnosed.

9.2 Policy implication

The results support the previous weakening of USPSTF guidelines and suggest that further weakening for asymptomatic women in their 40s could reduce overdiagnosis.

9.3 Takeaway

Clinical-trial behaviour is useful for guideline design because guidelines operate through behaviour, not only through average treatment effects.